gpeyt1-49 · Pile group block failure

GEO PE · Long Calculation Problems I — Topic 10: Deep Foundations

Problem

A 3 × 3 group of 12 in. square piles at 3 ft center-to-center spacing is driven 45 ft into soft-to-medium clay with weighted average su1 = 900 psf along the embedment and su2 = 1,600 psf below the toe (to 2B). Each single pile has nominal capacity 95 kips (α-method + toe). Pile cap is not in contact with the ground.

Most nearly, the nominal group capacity is:

A. 1,840 kips
B. 855 kips
C. 1,490 kips
D. 1,273 kips

Given equations / data

> Block: Rng = 2D(B + Z)su1 + B Z su2 Nc; Nc = 5(1 + D/5B)(1 + B/5Z) ≤ 9; B = Z = outside-to-outside group width; group capacity = lesser of block and Σ singles

Worked solution

1. B = Z = 2(3) + 1 = 7 ft; Nc = 5(1 + 45/35.0)(1.2) = 13.7 → cap at 9
2. Rblock = 2(45)(7.0+7.0)(900) + 7.0²(1,600)(9) = 1,839,600 lb
3. Σ singles = 9 × 95 = 855 kips
4. Group = lesser = 855 kips

Final answer

855 kips

Why the other values are wrong

- **1,840 kips** — took the larger value
- **1,490 kips** — used center-to-center (6 ft) group width for block
- **1,273 kips** — forgot the 2 on perimeter shear

References (Jayce's library)

- FHWA-NHI-16-009 (GEC 12) Vol I Eq. 7-36 and 7-37 (block failure, Nc)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=MI4gaUwboyE (no creator values copied).

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